# Azure Databricks

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| Project | Azure Databricks |
| Category | Data Engineering |
| Batch | batch_2 |

Purpose: Original educational notes created as a searchable PDF corpus for a portfolio-scale incremental RAG pipeline. The material is designed to contain meaningful technical sections that naturally produce multiple retrieval chunks.

## Databricks and the Lakehouse

Azure Databricks is a managed analytics platform built around Apache Spark and lakehouse concepts. It supports notebooks, SQL, jobs, workflows, governance, and Delta Lake. Data engineering workloads commonly ingest raw data, transform it through multiple quality layers, and expose curated tables for analytics.

**Practical note.** For data engineering work, the design should make processing boundaries explicit, preserve traceability, and avoid unnecessary recomputation when only a subset of the source has changed.

## Medallion Architecture

The medallion pattern organizes data into Bronze, Silver, and Gold layers. Bronze retains raw or minimally processed source data. Silver applies cleansing, standardization, deduplication, and business-valid transformations. Gold contains curated datasets such as dimensional models, aggregates, and business-facing tables. The layers are logical design concepts rather than mandatory product features.

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## Delta Lake

Delta Lake adds transaction-log-based reliability to data lake storage. Delta tables support ACID transactions, schema enforcement and evolution, updates, deletes, merges, and versioned table history. MERGE is especially useful for incremental pipelines that need to insert new records and update matching records.

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## Auto Loader and Structured Streaming

Auto Loader incrementally discovers new files in cloud object storage and is commonly used with Spark Structured Streaming. Checkpoints record streaming progress so a query can continue after restart without treating all previously processed input as new. Streaming does not necessarily mean millisecond processing; trigger configuration can also support incremental micro-batch patterns.

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## Unity Catalog

Unity Catalog provides centralized governance for data and AI assets. Its namespace commonly follows catalog.schema.object. Governance features include permissions, discovery, lineage, and managed access to tables, volumes, and other securable objects. Separating storage credentials and external locations can help control access to cloud storage.

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## Workflows and Jobs

Databricks Workflows can orchestrate notebook, Python, SQL, and other tasks with dependencies. A job can run on a schedule or be triggered manually or externally. Task dependencies make it possible to execute ingestion before transformations and validation. Operational design should avoid unnecessary recomputation and make failed tasks observable.

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